

Forecasting Financial Performance

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Forecasting Financial Performance

Chapter Overview

Valuing an ongoing business requires the same inputs as any other valuation exercise: (i) a forecast of the investment's free cash flow and (ii) an estimate of the appropriate risk-adjusted discount rate. However, with an ongoing business, there is an important distinction:

Firms typically have a history of past performance that is recorded in their financial statements, whereas proposed projects do not.

This history of a firm's financial statements contains a wealth of information that can be utilized in the valuation process. By using information from the financial statements about how the firm has made money in the past, we can forecast future free cash flows and future financial performance. However, before we can do this, we need to understand the relation between accounting earnings and cash flows. This is a primary objective of this chapter.

1 INTRODUCTION

This chapter provides our introduction to the problem of valuing an entire firm or business enterprise. Although the process used to value projects and entire firms is very similar, the valuation of a firm is in some ways easier because the firm has a history of past cash flows that can be used as a starting point for estimating future cash flows. But there's a catch. The firm's history is recorded in its financial statements, which are prepared in accordance with a set of accounting rules and guidelines that can differ significantly from cash flow. This means that we need to understand the construction of financial statements such that we can glean from them the firm's historical cash flows.

The focus of this chapter is on forecasting a firm's future cash flows. We begin our analysis of future cash flows by first analyzing the firm's historical financial statements. Consequently, it is crucial that we have a thorough understanding of the contents of a firm's financial statements and the interrelationships among them. We not only extract

information out of historical financial statements that helps us understand how the firm has made money in the past, but also use the financial statements as a model to forecast future performance.

We first review three of the four financial statements required of every public firm: the income statement, the balance sheet, and the statement of cash flow. The statement of retained earnings is the fourth required statement and is not germane to our purposes here. Next, we use planned, or pro forma, financial statements to model our forecast of future performance. Finally, we extract cash flows from the pro forma financial statements, which are in turn used to value the firm.

2 UNDERSTANDING FINANCIAL STATEMENTS AND CASH FLOW

Financial statements are prepared in accordance with a set of reporting guidelines called generally accepted accounting principles or GAAP. These guidelines come from a number of sources, including the accounting profession as well as various government regulatory bodies such as the Securities and Exchange Commission (SEC) and the Public Company Accounting Oversight Board (PCAOB). These rules and guidelines will not, in general, result in financial statements that provide information in the form that is most conducive for valuing a firm. Even so, as we illustrate in the pages that follow, a firm's financial statements can be recast in such a way as to reveal the firm's cash flows.

The Income Statement

The income statement is the first of three basic financial statements that we will discuss. Panel a of Table 1 contains the income statements for a hypothetical firm called Better Buys Inc. for 2008–2010 as they would be prepared by the firm's financial accountants and reported to the public. For example, in 2010, Better Buys recognized total revenues equal to \$37,219,250 and incurred costs of goods sold equal to \$24,192,513, leaving a gross profit of \$13,026,737. After subtracting operating expenses and depreciation expenses, its operating income is \$3,487,380.¹ To arrive at earnings before taxes, we must add the \$300,000 in nonoperating income and subtract the \$888,841 in interest expense, resulting in earnings before taxes of \$2,898,539. Subtracting a tax liability of \$1,159,416 produces net income after taxes of \$1,739,123 for 2010. Because the firm paid \$200,000 in dividends to its common stock holders \$1,539,123 was retained and reinvested in the firm.

Cash Versus Accrual Accounting—Net Income Is Not the Same As Cash Flow

The income statement is prepared using the principles of accrual, as opposed to cash accounting, which means that the entries in the income statement do not necessarily correspond to transfers of cash during the period for which the statement was prepared. This is a very important point, so let's dig a little deeper into what accrual accounting means for the preparation of the income statement.

¹ Depreciation and amortization expenses are not always separately identified in the income statement but are combined with operating expenses. When this occurs, one can identify the depreciation expense from the cash flow statement, which we discuss later.

Table 1 **Better Buys Inc. Income Statement****Panel a. Historical Income Statements**

	2008	2009	2010
Revenues	\$ 33,125,000	\$ 35,112,500	\$ 37,219,250
Cost of goods sold	(21,696,875)	(22,472,000)	(24,192,513)
Gross profit	\$ 11,428,125	\$ 12,640,500	\$ 13,026,737
Operating expenses			
Fixed cash operating expenses	(2,000,000)	(2,000,000)	(2,000,000)
Variable cash operating expenses	(5,300,000)	(5,618,000)	(6,029,519)
Depreciation expense	(1,343,750)	(1,424,375)	(1,509,838)
Net operating income	\$ 2,784,375	\$ 3,598,125	\$ 3,487,380
Nonoperating income	300,000	300,000	300,000
Earnings before interest and taxes	3,084,375	3,898,125	3,787,380
Interest expense	\$ (974,438)	\$ (860,855)	\$ (888,841)
Earnings before taxes	\$ 2,109,938	\$ 3,037,270	\$ 2,898,539
Taxes	(843,975)	(1,214,908)	(1,159,416)
Net income	\$ 1,265,963	\$ 1,822,362	\$ 1,739,123
Dividends paid	(200,000)	(200,000)	(200,000)
Addition to retained earnings	\$ 1,065,963	\$ 1,622,362	\$ 1,539,123

Panel b. Common Size Income Statements

	2008	2009	2010
Revenues	100.0%	100.0%	100.0%
Cost of goods sold	−65.5%	−64.0%	−65.0%
Gross profit	34.5%	36.0%	35.0%
Operating expenses			
Fixed cash operating expenses	−6.0%	−5.7%	−5.4%
Variable cash operating expenses	−16.0%	−16.0%	−16.2%
Depreciation expense	−4.1%	−4.1%	−4.1%
Net operating income	8.4%	10.2%	9.4%
Nonoperating income	0.9%	0.9%	0.8%
Earnings before interest and taxes	9.3%	11.1%	10.2%
Interest expense	−2.9%	−2.5%	−2.4%
Earnings before taxes	6.4%	8.7%	7.8%
Taxes	−2.5%	−3.5%	−3.1%
Net income	3.8%	5.2%	4.7%

The accounting income statement found in Panel a is prepared in accordance with generally accepted accounting principles (GAAP) and is based on the accrual method of accounting, which recognizes revenues in the period in which they are earned and the expenses incurred when generating those revenues. Panel b provides these same accounting numbers expressed as a percent of sales. These percentages are known collectively as the *common size income statements*. This common size statement is often used as a tool of financial statement analysis.

The accrual-based income statement is constructed using two fundamental principles: the *revenue recognition principle*, to determine which revenues to include in the statement, and the *matching principle*, to determine what expenses to include.

- *The revenue recognition principle and firm revenues.* According to the **revenue recognition principle**, revenue is recognized in the period in which products (goods or services), merchandise, or other assets are exchanged for cash or claims on cash. This may sound very academic, but revenue recognition is very important and it is sometimes difficult to determine when revenues should be reported as having been earned. For example, suppose a movie studio sells the rights to broadcast a movie it has not yet completed filming to one of the television networks. Should the revenue from the sale of the rights be reported when the contract is signed, when the movie is delivered to the network, when the network actually makes a cash payment to the studio, or when the movie is aired on the network? Most transactions are far less complicated than the sale of movie rights, but this example serves to demonstrate the difficulties that can arise when determining when a particular source of revenue should be reported in the firm's income statement.
- *The matching principle and expenses.* The expenses reported in the firm's income statement are those that were incurred in the process of generating the reported revenues. In other words, expenses are matched to reported revenues. This is the **matching principle**.

In summary, the firm's profits or net income for the period represent the difference in revenues recognized as earned during the period minus the matching expenses the firm incurred in generating those revenues, as shown in Equation 1 below.

$$\begin{array}{rcl} \text{Revenues} & - & \text{Expenses} \\ \text{for the Period} & & \text{for the Period} \end{array} = \text{Net Income} \quad (1)$$

Note that there is no requirement that cash ever change hands during the period. This means that reported net income for the period does not have to equal cash flow. To illustrate, consider the following example involving the purchase and sale of a big-screen television set:

- *On December 15, Better Buys Inc. orders and receives a new big-screen television set.* The set cost Better Buys \$2,500, but the manufacturer gave Better Buys credit terms of "net 90," such that the firm has 90 days to pay for the set with no penalty or interest charged.
- *On December 24, Better Buys Inc. sells the big-screen television for \$4,500.* The buyer opens a charge account with Better Buys for the entire purchase price, on which he will not have to make a payment for six months.
- *On December 31, Better Buys closes its books for the fiscal year just ended.* As part of its revenues for the year, Better Buys recognizes \$4,500 in revenues from the sale of the television and a corresponding cost of goods sold of \$2,500.

Better Buys does not collect any cash for six months, nor does the firm pay for the big-screen television for 90 days, however, the firm's income statement for the period will report additional profits before taxes of \$2,000!

Operating Versus Nonoperating Income

Another important feature of the way in which the accounting profession presents a firm's income statement is the separation of operating and nonoperating income. **Operating income**, as the name suggests, is the income generated by the firm's core business operations. For Better

Buys Inc., this would be the sale of TVs and other electronic appliances that generated an operating profit of \$3,487,380 for 2010. **Nonoperating income** is generated by investments the firm has made in assets that are unrelated to the firm's primary business. For example, gains from the sale of investments or the sale of a subsidiary or division would generally be recorded as nonoperating income. Other examples include the costs incurred from restructuring, currency exchange, or the write-off of obsolescent inventory. In 2010, Better Buys earned \$300,000 in nonoperating income from \$3 million it has invested in other businesses.

Common Size Income Statement

Panel b of Table 1 contains common size income statements for Better Buys Inc. spanning the period 2008–2010. These statements are simply standardized income statements in which each line item is converted to a percent of revenues for the year. Three line items in the common size income statement entries are highlighted using block arrows: Gross profit/Revenues, Net operating income/Revenues, and Net income/Revenues. These ratios are collectively referred to as **profit margins**—specifically the **gross profit margin**, **operating profit margin**, and **net profit margin**, respectively.

Analysts pay close attention to a firm's profit margins because they convey important information about the past performance of a firm. For example, in 2010, Better Buys Inc. had 35% of each dollar of revenues left over after paying for the goods and services it sold. The firm also kept 9.4% out of each dollar of sales after paying all the firm's operating expenses. Finally, after including the firm's sources of nonoperating income, paying interest expenses, and recognizing the firm's liability for the year, we see that Better Buys earned net income after taxes equal to 4.7% of revenues.

To evaluate these profit margins, analysts typically compare them to those of similar firms in the industry and examine whether they are improving or deteriorating over time. Figure 1 contains sample data on profit margins for three consumer electronics firms—Best Buy Inc. (BBY), GameStop Corp. (GME), and Conn's Inc. (CONN)—as well as for retail giant WalMart (WMT). The profit margins are all very similar, suggesting that gross profits are about 25% of firm revenues, operating profits are about 7% of revenues, and net profits are about 4% of revenues. These profit margins reflect the operations of a retail firm. The numbers are somewhat different for automaker Toyota (TM), which has a lower gross profit margin but higher operating and net profit margins. The point we are trying to make is that different types of businesses have unique profitability attributes. So, when you analyze the historical performance of a firm, you should look for anomalies that suggest that the firm is a particularly good performer or perhaps a laggard.

Panel b of Figure 1 presents trailing 12-month profit margins for Best Buy Inc., the computers and electronics industry, and all firms in the S&P 500 index. Note first that Best Buy has performed very well in the most recent years when compared to others in its industry. However, the profit margins in this industry are much lower than those of the average firm found in the S&P 500. This latter point further emphasizes that comparisons of financial performance must be restricted to firms that operate in similar environments and face similar competitive forces.

The Balance Sheet

The balance sheet provides a snapshot of the firm's financial position at a moment in time and a detailed account of the company's assets, liabilities (debts), and shareholders' equity. The balance sheet equation, shown in Equation 2, requires that the sum of the book values of the firm's assets equal the sum of the debts the firm owes to its creditors plus the investment of the stockholders' equity.

Figure 1 Profit Margins for Three Consumer Electronics Firms, WalMart (WMT), and Toyota (TM)**Panel a. Profit Margin Data for 2007–2008**

	Best Buy Inc. (BBY)		GameStop Corp. (GME)		Conn's Inc. (CONN)		WalMart (WMT)	
	2007	2008	2007	2008	2007	2008	2007	2008
Gross profit margin	24.4%	23.9%	27.7%	25.6%	26.8%	25.9%	23.4%	23.2%
Operating profit margin	5.6%	5.4%	6.2%	6.9%	8.0%	7.2%	5.6%	5.8%
Net profit margin	3.8%	3.5%	3.0%	4.1%	5.3%	4.8%	3.3%	3.4%

	Toyota Motor Corp. (TM)	
	2007	2008
Gross profit margin	19.7%	18.1%
Operating profit margin	9.3%	8.6%
Net profit margin	6.9%	6.5%

Panel b. Best Buy Inc., Industry Average, and S&P 500 Average Profit Margin Ratios

	Best Buy Inc.	Industry	S&P 500
Gross margin (TTM)	24.14%	1.16%	41.28%
Gross margin—5 Yr. Avg.	24.19%	20.9%	40.68%
Operating margin (TTM)	4.4%	0.19%	—
Net profit margin (TTM)	2.69%	0.1%	9.88%

Legend: TTM = trailing 12 months.Source: Reuters (<http://www.reuters.com/finance/stocks/ratios?symbol=BBY.N&rpc=66profitability>)

Profit margins for 2007–2008 include the following: Gross profit margin = Gross profit/Sales, Operating profit margin = Net operating income/Sales, and Net profit margin = Net income/Sales.

$$\text{Assets} = \text{Liabilities} + \text{Stockholders' Equity} \quad (2)$$

The account balances shown in the balance sheet change over time to reflect the action taken by the firm's management. As such, the balance sheet contains a history of the firm's decisions regarding the specific assets in which the firm has invested, how the firm has financed those assets (debt versus equity), how profitable the firm has been over time (the size of the firm's retained earnings balance), and the firm's capital intensity (the relative size of the firm's investments in long-term or fixed assets compared to the size of total assets). The assets of the firm are divided into current assets and long-term assets (net property, plant, and equipment plus investments), and the financing side of the balance sheet is broken down into liabilities and owners' equity. The liabilities are further divided into current liabilities that are due and payable within a period of one year or less and long-term liabilities. Four years of Better Buys Inc.'s historical balance sheets are found in Panel a of Table 2.

Table 2 **Better Buys Inc. Balance Sheets****Panel a. Historical Balance Sheets**

Balance Sheets	2007	2008	2009	2010
Cash and marketable securities	\$ 624,000	\$ 650,000	\$ 702,500	\$ 744,385
Accounts receivable	6,651,500	7,102,000	7,408,738	8,113,797
Inventories	10,001,000	10,643,000	11,938,250	12,282,353
Current assets	\$ 17,276,500	\$ 18,395,000	\$ 20,049,488	\$ 21,140,534
Net property, plant, & equipment	13,437,500	14,243,750	15,098,375	16,004,278
Investments	3,000,000	3,000,000	3,000,000	3,000,000
Total	\$ 33,714,000	\$ 35,638,750	\$ 38,147,863	\$ 40,144,812
Accounts payable	\$ 7,812,500	\$ 8,281,250	\$ 8,778,125	\$ 8,800,100
Short-term debt	6,875,000	7,287,500	7,724,750	8,188,235
Current liabilities	\$ 14,687,500	\$ 15,568,750	\$ 16,502,875	\$ 16,988,335
Long-term liabilities	4,603,061	4,563,719	4,504,195	4,464,966
Total liabilities	\$ 19,290,561	\$ 20,132,469	\$ 21,007,070	\$ 21,453,301
Deferred income taxes	\$ 329,749	\$ 346,629	\$ 358,778	\$ 370,372
Common stock	\$ 10,000,000	\$ 10,000,000	\$ 10,000,000	\$ 10,000,000
Retained earnings	4,093,690	5,159,653	6,782,015	8,321,139
Total owners' equity	14,093,690	15,159,653	16,782,015	18,321,139
Total liabilities and owners' equity	\$ 33,714,000	\$ 35,638,750	\$ 38,147,863	\$ 40,144,812

Panel b. Common Size Balance Sheets (standardized by total assets for each year)

	2007	2008	2009	2010
Cash and marketable securities	2%	2%	2%	2%
Accounts receivable	20%	20%	19%	20%
Inventories	30%	30%	31%	31%
Current assets	51%	52%	53%	53%
Net property, plant, & equipment	40%	40%	40%	40%
Investments	9%	8%	8%	7%
Total	100%	100%	100%	100%
Accounts payable	23%	23%	23%	22%
Short-term debt	20%	20%	20%	20%
Current liabilities	44%	44%	43%	42%
Long-term liabilities	14%	13%	12%	11%
Total liabilities	57%	56%	55%	53%
Deferred income taxes	1%	1%	1%	1%
Common stock	30%	28%	26%	25%
Retained earnings	12%	14%	18%	21%
Total owners' equity	42%	43%	44%	46%
Total liabilities and owners' equity	100%	100%	100%	100%

The balance sheet represents a summary of a company's financial condition at a point in time. The balance sheet equation states that the sum of the firm's assets must equal the sum of the sources of financing used to acquire those assets (stockholders' equity plus the firm's debts). Panel a includes historical balance sheet data for 2007–2010. Panel b contains the common size balance sheet accounts, where each line item in the balance sheet is divided by total assets for the year.

Measurement—Where Do the Numbers Come From?

Historical cost is the primary guiding principle underlying the construction of a firm's balance sheet. Very simply, when a firm acquires an asset or incurs a liability, the amount recorded on its balance sheet is the dollar value of the transaction *at the time*. However, following this principle does not mean that the account balance remains equal to the historical cost forever. As Figure 2 illustrates, balance sheet entries for some types of accounts are adjusted to current market values on a regular basis using a process called mark-to-market accounting, while others are adjusted only in the event that the market value of the asset declines. Understanding the accounting practices underlying the determination of the account balances is essential to gaining a full appreciation of the firm's financial condition depicted in the balance sheet.

Nonoperating Assets and Investments

Better Buys has a line item titled “Investments,” which has a \$3 million balance. This line item (sometimes referred to as long-term investments because they are assets the company intends to hold for longer than one year) can consist of stocks and bonds of other companies, real estate, as well as cash set aside for a specific purpose. An equity investment involving more than 20% but less than 50% of the equity of another firm is recorded as an investment and is titled something like “Investments in unconsolidated subsidiaries.” If the firm owns more than 50% of a subsidiary, then the rules of accounting call for consolidation of the subsidiary into the firm's balance sheet and income statement. When this occurs, a minority interest account appears between liabilities and owners' equity, representing the value of the minority-owned shares, because 100% of the value of the subsidiary's assets were incorporated into the consolidated balance sheet.

The important thing to note here is that the presence of a long-term investment category on a firm's balance sheet signifies the presence of a potentially significant nonoperating asset that will need to be valued. This means that the analyst will need to dig deeper into the footnotes of the financial statement to determine the exact nature of the investments in order to determine an appropriate approach to valuing them. For example, in 2008, the Starbucks Corporation (SBUX) had the following investments listed in its balance sheet:

Long-term investments—available-for-sale securities	\$ 71,400,000
Equity and cost investments	<u>302,600,000</u>
Total Investments	\$374,000,000

Furthermore, from Note 4 of Starbucks 2008 financial statements, we learn that the long-term available-for-sale investments included \$59.8 million of illiquid auction rate securities. The auction market failures that began in mid-February 2008 made these securities illiquid, so they were reclassified as long-term investments (they were previously classified as short-term available-for-sale securities). Note 7 reveals that Starbucks' Equity and Cost Investments totaling \$302.6 million were comprised of \$267.9 million in equity method investments at the end of 2008 and \$34.7 million in cost method investments (see Figure 2 for a discussion of these categories). The equity method investments consisted of various limited partnerships (StarCon LLC, which is a joint venture formed in March 2007 with Concord Music Group Inc., and various international Starbucks affiliates from countries around the world). The investments recorded using the cost method

Figure 2 **The Balance Sheet—How are the account balances determined?**

Assets	Definition	Method of Measurement
Cash and cash equivalents	These are the most liquid assets on the firm's balance sheet. These include deposits that mature in 90 days or less.	Fair value equal to the amount of cash held.
Short-term investments	Investments in short-term government bonds, marketable securities, and commercial paper.	Market value, with the exception of long-term securities that are being held to maturity (see long-term debt securities below).
Receivables	The amount owed to the company by its customers who made credit purchases.	The amount of cash the firm expects to collect from its customers from prior sales.
Inventories	The goods and materials that a firm holds in stock.	Lower of cost or market value.
Long-term tangible assets	Property, plant, and equipment that are used in the firm's principal line of business.	Historical cost less accumulated depreciation unless the fair value of the assets drops below this amount, in which case they are written down and a charge to earnings is taken to reflect the write-down.
Intangible assets	Identifiable nonmonetary assets that cannot be seen, touched, or physically measured. Examples include copyrights, patents, trademarks, and goodwill.	Historical cost. If there is a defined life of the right, then its value is written off over time. Otherwise, the recorded value is changed only if its market value declines below the recorded or carrying value on the firm's balance sheet.
Goodwill	An intangible asset that arises when a firm purchases another firm and the purchase price exceeds the acquired firm's fair value, where fair value is the amount at which the asset could be bought or sold in a current transaction between willing parties or transferred to an equivalent party, other than in a liquidation sale.	Historical cost. Should the value of the acquired firm's assets decline after the acquisition, then goodwill is written off; otherwise, it remains at historical cost.
Investments—Long-term debt securities	Debt securities such as bonds that are held as investments by the firm.	Three categories are used to determine the recorded value: (1) <i>Investments held for active trading</i> —fair market value with adjustments for unrealized gains and losses (market to market); (2) <i>Investments available for sale</i> (not for trading but may be sold before maturity)—recorded at fair market value, but unrealized gains and losses do not affect the asset balance, instead resulting in adjustments to equity, and interest income is reported in the income statement; and (3) <i>Investments held to maturity</i> —recorded at historical cost with no adjustment for unrealized gains and losses, but interest income is recorded in the income statement.

(continued)

Figure 2 continued

Assets	Definition	Method of Measurement
Investments—Equity	Equity securities purchased and held by the firm as an investment.	The method for determining the reporting of these investments depends on the fraction of the equity of the shares of the issuing firm that the investing company owns: (1) Less than 20% ownership—these investments are classified using the same three categories used to categorize debt security investments and accounted for on the balance sheet in the same way; (2) Investments involving more than 20% but less than 50% ownership—recorded at historical cost of the investment but written up to record the firm's share of the earnings of the investment company and down for dividends paid or goodwill that is written off (the share of the firm's earnings and write-off of goodwill is also recorded in the firm's income statement); (3) Investments involving more than 50% of the equity of another firm—the financial statements of the two firms are consolidated, with a deduction for minority interests in the net assets (in the balance sheet) and net income (in the income statement).
Liabilities	Definition	Method of Measurement
Payables (short-term)	Liabilities incurred in the normal course of business as a result of the granting of credit. Examples include accounts payable, interest payable, and taxes payable.	Fair value, which equals the amount of cash required to satisfy the obligation.
Short-term debt and long-term debt	Financial obligations arising out of the firm's borrowing money that is to be repaid in a period of one year or less is short-term. Examples include bank loans and lease obligations. Debt obligations that do not mature within a period of one year are long-term.	Fair value, which is equal to the present value of the contractual payment obligations. This represents the market value of the obligations at the time they are recorded; however, as interest rates change over time, the changes in the values of these obligations are not adjusted. However, the debt footnote to the balance sheet typically compares market values of these obligations to their book values.
Accrued and estimated liabilities	Liabilities that arise in the firm's operations, including pension liabilities, warranty liabilities, and estimated restructuring liabilities.	"Quasi" fair value, which is estimated using the present value of the expected future financial obligation. The term <i>quasi</i> reflects that fact that there are no market values available to use when adjusting the value of these liabilities and, thus, the amounts used are estimated. Because the account value is adjusted to an estimated value, this practice is sometimes euphemistically referred to as market-to-model accounting.

investments represent equity interests in entities to develop and operate Starbucks-licensed retail stores in several global markets including Mexico, Hong Kong, and Greece. Starbucks' equity investments in unrelated businesses was only \$2.8 million.

The point we are making is that the "Other assets" or "Investments" line items on the firm's balance sheet can contain significant assets whose values will not be captured when we calculate the present value of the firm's estimated free cash flow, this happens because "Other income" is not included in firm free cash flow. This, in turn, means that we will need to value these investments independently of the firm's operating assets and add them to our discounted firm free cash flow calculation.

The Common Size Balance Sheet

Panel b of Table 2 contains **common size balance sheets**, which are constructed by dividing each balance sheet line item by total assets. The common size balance sheet is a popular tool of financial statement analysis, for it provides a quick answer to some very basic questions about the firm. For example, on the financing side of the balance sheet, we see that 53% of Better Buys' total assets in 2010 were financed by debt. Of these total liabilities, 42% are current liabilities that must be repaid in a year or less. This percentage suggests that the firm could find itself unable to repay its debts in a timely fashion and thus be forced into financial insolvency. It is not possible to say that the firm faces an unusual degree of financial risk based on this one ratio alone. However, the analyst might use this indicator as the motivation for exploring firm liquidity and financial risk more fully.

Statement of Cash Flow

The **statement of cash flow** (sometimes referred to as the **cash flow statement**) reports a company's inflows and outflows of cash. This statement converts information from the firm's income statement and balance sheet, which are both prepared using the principles of accrual accounting, into cash flows for the period.

Table 3 contains Better Buys Inc.'s statement of cash flow for 2010. The statement details the sources of cash inflow to the firm and how the firm spent cash for the period, which when summed equals the change in the firm's cash balance for the period. Specifically, the cash flow statement is divided into three parts that represent the firm's cash flows from (1) operating activities, (2) investing activities, and (3) financing activities. Thus, in equation form, the cash flow statement is as follows:

$$\left(\begin{array}{c} \text{Cash Flows from} \\ \text{Operating Activities} \end{array} \right) + \left(\begin{array}{c} \text{Cash Flows from} \\ \text{Investing Activities} \end{array} \right) + \left(\begin{array}{c} \text{Cash Flows from} \\ \text{Financing Activities} \end{array} \right) = \begin{array}{c} \text{Change in Cash Balance} \\ \text{for the Period} \end{array} \quad (3)$$

For 2010, Better Buys experienced a \$41,885 increase in its cash balance that resulted from a \$2,233,369 cash inflow from the firm's operations, a (\$2,415,740) cash outflow for its investing activities, and a \$224,256 cash inflow from financing activities.

Free Cash Flow Computations

Our discussion to this point has focused on the firm's financial statements that are prepared using the rules of GAAP. It is now time to switch gears and focus on the free cash flow generated by the firm during 2010. There are two definitions of free cash flow: firm (or project) free cash flow and equity free cash flow. Both these measures of free cash flow can be computed from information contained in the firm's financial statements.

Table 3 Better Buys Inc. Statement of Cash Flow for 2008–2010

	2008	2009	2010
<i>Cash Flows from Operations</i>			
Net income	\$ 1,265,963	\$ 1,822,362	\$ 1,739,124
Depreciation	1,343,750	1,424,375	1,509,838
	<u>\$ 2,609,713</u>	<u>\$ 3,246,737</u>	<u>\$ 3,248,962</u>
(Increase) Decrease in accounts receivable	(450,500)	(306,738)	(705,059)
(Increase) Decrease in inventory	(642,000)	(1,295,250)	(344,103)
	<u>\$ (1,092,500)</u>	<u>\$ (1,601,988)</u>	<u>\$ (1,049,162)</u>
Increase (Decrease) in accounts payable	\$ 468,750	\$ 496,875	\$ 21,975
Increase (Decrease) in deferred income taxes	16,880	12,149	11,594
<i>Cash provided by operations</i>	<u>\$ 2,002,842</u>	<u>\$ 2,153,774</u>	<u>\$ 2,233,369</u>
<i>Cash Flows from Investing Activities</i>			
(Increase) Decrease in net fixed assets	(2,150,000)	(2,279,000)	(2,415,740)
<i>Cash flows from investing activities</i>	<u>(2,150,000)</u>	<u>(2,279,000)</u>	<u>(2,415,740)</u>
<i>Cash Flows from Financing Activities</i>			
Increase (Decrease) in short-term debt	\$ 412,500	\$ 437,250	\$ 463,485
Increase (Decrease) in long-term liabilities	(39,342)	(59,524)	(39,229)
Increase (Decrease) in common stock	—	—	—
Dividends paid	(200,000)	(200,000)	(200,000)
<i>Cash flows from financing activities</i>	<u>\$ 173,158</u>	<u>\$ 177,726</u>	<u>\$ 224,256</u>
<i>Change in Cash</i>	<u>\$ 26,000</u>	<u>\$ 52,500</u>	<u>\$ 41,885</u>

The statement of cash flow explains the change in cash balance for the period. It does so by decomposing firm cash flows into three categories: operating activities, investment activities, and financing activities. For example, Table 2 shows that at the end of 2009, Better Buys Inc. had a cash balance of \$702,500, and at the end of 2010, this balance was \$744,385, for a change in cash balance over the period of \$41,885.

Firm (Project) Free Cash Flow

Panel a of Table 4 contains estimates of the 2010 firm free cash flow (Firm FCF) for Better Buys Inc., prepared using the direct and indirect methods. The direct method focuses directly on the cash flows generated by the firm's operations, whereas the indirect method focuses on the distribution of those cash flows to the firm's investors.

Direct method for calculating firm free cash flow The direct method is the methodology we have been using throughout the book to calculate free cash flow. It begins by

Calculating CAPEX for 2010

Better Buys Inc. invested an additional \$2,415,740 in new plant and equipment during 2010. We can verify this capital expenditure by examining the change in net property, plant, and equipment between 2010 and 2009:

Net property, plant, and equipment (2009)	\$ 15,098,375
Less: Depreciation expense (2010)	<u>(1,509,838)</u>
Plus: CAPEX (2010)	<u>2,415,740</u>
Net property, plant, and equipment (2010)	\$ 16,004,278

Calculating operating net working capital for 2010

Better Buys Inc. invests an additional \$1,157,613 in 2010, which is calculated as the change in operating net working capital between 2010 and 2009 as follows:

	2009	2010	Change
Cash and marketable securities	\$ 650,000	\$ 702,500	\$ 52,500
Accounts receivable	7,102,000	7,408,738	305,735
Inventories	10,643,000	11,938,250	1,295,250
Current assets	\$ 18,305,000	\$ 20,049,488	\$ 1,654,488
Less: Accounts payable	(8,281,250)	(8,778,125)	\$ (490,875)
Equals: Operating net working capital	\$ 10,113,750	\$ 11,271,303	\$ 1,157,613

Thus, the net new investment in operating net working capital is equal to the change in the firm's levels of operating net working capital for the period.

Note that in this situation, we assume that cash and marketable securities are part of the firm's operating current assets. As such, cash and marketable securities are included in the calculation of operating net working capital. This is not always the case in practice, where these accounts are not considered operating assets and are omitted from the working capital calculation.

calculating net operating profit after tax, adds back noncash expenses such as depreciation for the period plus any adjustment to deferred taxes, then subtracts capital expenditures (CAPEX) and any increase in net operating working capital to get a Firm FCF for 2010.

The adjustment to NOPAT for the change in the firm's **deferred tax liability** deserves some explanation because it did not arise earlier when we calculated project free cash flow. Note that the tax liability calculated using the firm's net operating income that we referred to as "taxes on operating income" is simply the firm's current tax rate(s) applied to the firm's operating income. However, because firms often pay taxes on income computed in a slightly different way than the income they report (i.e., they use accelerated depreciation to calculate their tax liability but use straight-line depreciation for financial reporting), they often pay less in taxes than is reported in the firm's income statement. The difference accumulates in a deferred income tax liability, which will eventually be drawn down as the firm pays the tax it

owes in the future. To adjust for the difference in taxes actually paid and the tax liability incurred by the firm during the period, we add back any increase in the deferred income tax liability and subtract any decrease in this account when calculating the firm's free cash flow.

Indirect method for calculating firm free cash flow The indirect method for calculating firm free cash flow focuses on how the firm's cash flow gets distributed among its creditors and investors. Our focus has been, and will continue to be, on the direct method; however, the indirect method provides a useful reminder of just what Firm FCF represents (i.e., the cash available for distribution to the firm's creditors and investors). For example, in 2010, Better Buys Inc. distributed \$533,305 after taxes to its short- and long-term creditors in the form of interest payments, obtained an additional \$463,485 from its short-term creditors in the form of added loans, distributed \$39,229 to its long-term creditors in principal repayments, and paid \$200,000 to its common stockholders in cash dividends.

Other income and firm free cash flow Better Buys Inc.'s Firm FCF available for distribution to all its investors for 2010 was \$309,049. Note that this total includes both the free cash flow from operations and the after-tax nonoperating income. This raises an important point about firm compared to project valuation. Firms often earn income from

Table 4 Better Buys Inc. Firm and Equity Free Cash Flow Calculations for 2008–2010

Panel a. Firm Free Cash Flow (Firm FCF) Calculation	2008	2009	2010
Direct Method			
Net operating income	\$ 2,784,375	\$ 3,598,125	\$ 3,487,382
Less: Taxes on operating income	(1,113,750)	(1,439,250)	(1,394,953)
Net operating profit after taxes (NOPAT)	\$ 1,670,625	\$ 2,158,875	\$ 2,092,429
Increase (Decrease) deferred taxes	\$ 16,880	\$ 12,149	\$ 11,594
Plus: Depreciation	1,343,750	1,424,375	1,509,838
Less: CAPEX	(2,150,000)	(2,279,000)	(2,415,740)
Less: Change in operating net working capital	(649,750)	(1,157,613)	(1,069,072)
Firm free cash flow (from operations)	\$ 231,505	\$ 158,787	\$ 129,049
Nonoperating income after taxes at 40%	180,000	180,000	180,000
Firm free cash flow available for all investors	\$ 411,505	\$ 338,787	\$ 309,049
Indirect Method			
Interest (1-Tax Rate)	\$ 584,663	\$ 516,513	\$ 533,305
(Increase) Decrease in short-term debt	(412,500)	(437,250)	(463,485)
(Increase) Decrease long-term liabilities	39,342	59,524	39,229
Dividends	200,000	200,000	200,000
(Increase) Decrease in common stock	—	—	—
Firm free cash flow available for all investors	\$ 411,505	\$ 338,787	\$ 309,049
Panel b. Equity Free Cash Flow (Equity FCF) Calculation			
Direct Method			
Net income	\$ 1,265,963	\$ 1,822,362	\$ 1,739,124
Plus: Depreciation	1,343,750	1,424,375	1,509,838
Increase (Decrease) in long-term liabilities	(39,342)	(59,524)	(39,229)
Increase (Decrease) in short-term debt	412,500	437,250	463,485
Increase (Decrease) in deferred income taxes	16,880	12,149	11,594
Less: Change in operating NWC	(649,750)	(1,157,613)	(1,069,072)
Less: Capital expenditures (CAPEX)	(2,150,000)	(2,279,000)	(2,415,740)
Equity FCF	\$ 200,000	\$ 200,000	\$ 200,000
Indirect Method			
Dividends	200,000	200,000	200,000
Less: New stock issues	—	—	—
Plus: Stock repurchases	—	—	—
Equity free cash flow (Equity FCF)	200,000	200,000	200,000
Firm and equity free cash flows for 2008–2010 are calculated using both the direct and indirect methods.			

both operating and nonoperating asset investments. The first, as we discussed earlier when describing Better Buys Inc.'s balance sheet, arises out of its core business operations. For example, Better Buys earned a free cash flow of \$129,049 from its operations in 2010. In addition, firms often earn other income from investments they have made in unrelated business activities or from interest income on cash and marketable security accounts. For example, in 2010, Better Buys earned an after-tax amount of $\$180,000 = \$300,000 \times (1 - .40)$ from such investments. These nonoperating cash flows, like operating cash flows, are a source of value for the firm and must be incorporated into our analysis. Standard practice involves separating these two categories of income and valuing them independently.

Nonoperating income can also arise out of excess capacity in operating assets that the firm might hold. For example, if the firm does not need all its warehouse space to support the projected level of operating income and if that excess capacity can be leased to another firm, then the rental income from leasing the excess capacity forms a source of nonoperating income. Excess capacity can also arise out of the one-time sale of plant and equipment the firm owns but does not need for supporting its operations. This type of other income is often an important motivation for the acquisition of one firm by another.

One final source of "other income" from excess capacity comes in the form of excess cash and marketable security holdings by the firm. In fact, some valuation analysts consider 100% of a firm's cash holdings as excess in the sense that they are not needed to support the firm's operations. This is an extreme position, and a more reasonable rule of thumb that is sometimes used is to assume that the firm needs a small percent, say, 2%, of its revenues in cash and marketable securities and that any excess is considered a source of other income. For example, if a firm's cash balance were \$10 million and 2% of revenues were only \$1 million, then an acquiring firm might consider that by acquiring the firm, they would gain an asset worth \$9 million that would need to be added to the value of the firm's operating assets.

Equity Free Cash Flow

Panel b of Table 4 includes both the direct and indirect methods of estimating equity free cash flow (Equity FCF), which equals \$200,000 for 2010. Note that the direct method begins with net income, because Equity FCF focuses only on the cash flow available for distribution to the firm's common shareholders. Net income is then adjusted for the effects of noncash depreciation, changes in the firm's short- and long-term debt, deferred taxes, and net new investments in both capital equipment and operating working capital.

The indirect method focuses on the distribution of cash to the firm's shareholders through the payment of cash dividends or share repurchases net of any new equity financing the firm might have obtained from the sale of shares. In 2010, Better Buys Inc. did not repurchase or issue any shares of stock but did pay a cash dividend of \$200,000, so the Equity FCF for 2010 was \$200,000.

Equity FCF is the appropriate measure of cash flow to use when directly valuing the firm's equity in a project.

Reconciling the Cash Flow Statement with Firm FCF

The firm's cash flow statement in Table 3 contains a line item titled "Cash provided by operations," which was equal to \$2,233,369 for 2010. This term sounds very much like "Firm free cash flow (from operations)" that we calculated using the direct method for estimating Firm FCF in Table 4. These two quantities are obviously not the same,

because we calculated free cash flow from operations for Better Buys to be only \$129,049. Why is cash flow from operations found in the cash flow statement not equal to firm free cash flow (from operations)? Exploring the reasons for the difference in these two quantities gives us an opportunity to develop the link between the firm's cash flow statement and firm free cash flow. This is important, for in this analysis we demonstrate that the FCF calculation we have been using to value projects in earlier chapters and will use to value the firm in subsequent chapters is consistent with the firm's accounting statements. We will rely upon this relationship later when we prepare our financial forecasts using the format of the firm's financial statements using pro forma statements.

There are three reasons for this difference between the line item in the cash flow statement titled "Cash provided by operations" and the firm's free cash flow (from operations): Specifically, cash provided by operations from the cash flow statement:

1. is measured after subtracting after-tax interest expense, because interest expense is deducted in the calculation of net income (whereas FCF is calculated before the deduction of interest expense),
2. includes after-tax nonoperating or other income, because this source of income is also part of net income (whereas FCF separates cash flow from operations and nonoperating activities), and
3. excludes consideration for capital expenditures, because these expenditures are included in the cash flow statement in a separate line item called *Cash Flows from Investing Activities*.

To reconcile the cash provided by operations with firm free cash flow (from operations), we begin with cash provided by operations from the cash flow statement. Next we subtract the increase in cash balance of \$41,885, which is included in the estimate of the change in operating net working capital, add back after-tax interest expense, subtract after-tax sources of nonoperating or other income, and finally subtract capital expenditures:

Cash provided by operations	Cash flow statement	\$2,233,369
Less: Increase in cash balance	Cash flow statement	(41,885)
Plus: After-tax interest expense ²	\$888,841 (1 – .40)	533,305
Less: After-tax other income	\$300,000 (1 – .40)	(180,000)
Less: Capital expenditures (CAPEX)	Cash flow statement	(2,415,740)
Firm free cash flow (from operations)		<u>\$129,049</u>

So, even though the line item found in the firm's cash flow statement captioned "Cash provided by operations" sounds very similar to the "Firm free cash flow (from operations)," they are not the same thing; however, they are related, as the above reconciliation demonstrates.

Free Cash Flow and Nonoperating Income

We now focus our attention on something we referred to simply as firm free cash flow. In Table 4 where we calculate the firm free cash flow for the Better Buys example, we identify

² Recall that the corporate tax rate was 40% in the Better Buys example.

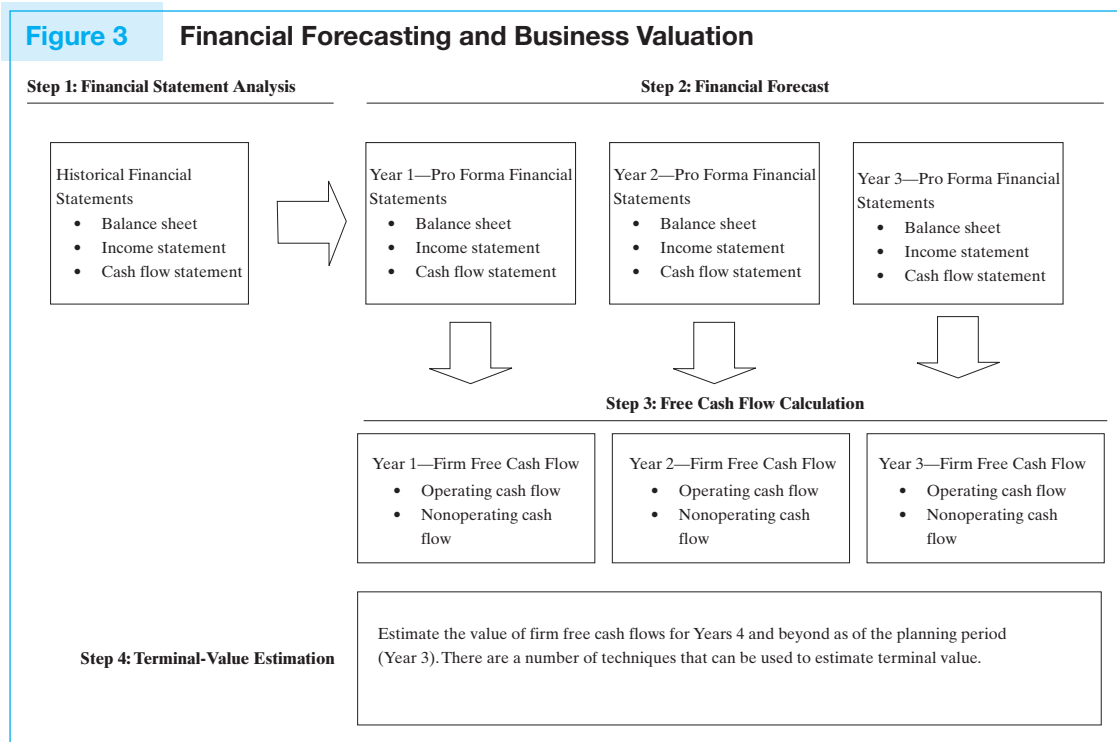
two similar-sounding free cash flow numbers: firm free cash flow (from operations), equal to \$129,049 for 2010, and free cash flow available for investors, which equals \$309,049. The difference in these two cash flow numbers is equal to the \$180,000 that is the after-tax nonoperating income that Better Buys Inc. earned during 2010.

Nonoperating income is income derived from activities that are unrelated to the firm's core business operations. Examples would include such things as dividend income, profits and losses from investments, gains and losses from foreign exchange, and capital gains resulting from the sale of real estate investments that were not related to the firm's operations. The amount of nonoperating income will vary from one firm to another and over time as well. Because the determinants of operating and nonoperating income are different, standard practice involves making separate analyses and forecasts for each. We return to this point later when we discuss forecasting.

3 FORECASTING FUTURE FINANCIAL PERFORMANCE

Now that we have reviewed the content of a firm's financial statements and reconciled them to the cash flow estimate that we need to support our effort to value a business, it is time to take a closer look at financial forecasting. We can think of the financial forecast as a four-step process, which we describe below and illustrate in Figure 3:

Step 1: *Perform an analysis of the firm's most recent financial statements.* Our ability to develop an understanding of how a firm will make money in the future is greatly enhanced by a thorough understanding of how it has made money in the past.



- Step 2:** *Construct pro forma financial statements for a three-to five-year planning period.* Accounting financial statements are the standard model used to evaluate a firm's past and projected future performance.
- Step 3:** *Convert pro forma financial statement forecasts to predictions of the firm's free cash flow from operations and free cash flow from nonoperating income.* As we discussed in the previous section, we can extract the necessary information for a forecast of future cash flow from a firm's income statement and balance sheet.
- Step 4:** *Forecast the value of cash flows after the end of the planning period using a terminal value.* Although detailed pro forma financial statements could be prepared for an extended period of time (beyond the three to five years noted in step 2), analysts typically condense the valuation of post-planning-period cash flows into a single terminal-value estimate.

Step 1: Perform an Analysis of Historical Financial Statements

There are a number of data sources for financial statements. The firm's annual report or 10K filing with the SEC is provided by the SEC at <http://www.sec.gov/edgar.shtml>. The data are also available in a form that is somewhat easier to access from Standard & Poor's Compustat data service (<https://www.compustatresources.com/support/index.html>). In addition, there are free database sources such as those found at Yahoo (<http://finance.yahoo.com/?u>) or Google (<http://finance.yahoo.com/?u>). There is, however, a downside to the standardization of financial information found in these large common databases. In order to develop a common format and eliminate the peculiarities of the reporting formats of individual companies, some data entries have to be combined into general categories such as "other assets" or "other liabilities." This practice can sometimes obscure important assets and liabilities and make them harder to forecast, thereby creating forecasting errors.

Once the historical financial statements have been collected, the next phase in the analysis involves looking for relationships between the individual line items in the statements and firm revenues, which can be used to build our forecast of pro forma financial statements. Good places to start this analysis are the common size income statement and the balance sheet, which we saw earlier in Tables 1 and 2. If we have multiple years of historical financial statements, then we can use the common size ratios (e.g., Cost of goods sold/Revenues) to look for stable relationships that can be used to forecast the individual line items in the income statement and balance sheet. For example, look at Panel b of Table 1 and the gross profit margin identified by the top arrow. The gross profit margin, which is the percent of firm revenues left after paying for cost of goods sold, appears to vary randomly about its average of 35.16%. Thus, in the absence of any information about the size of the firm's cost of goods sold, we might forecast gross profits in the future to equal the average of the past three years.

The term *financial statement analysis* refers to the process of analyzing a firm's financial statements to learn about how well the firm is being managed and the firm's future performance potential. In the pages that follow, we provide a brief overview of the basic elements of financial statement analysis. However, before we embark, we first point out some important observations concerning a firm's financial statements that should be kept in mind along the way:

1. *Financial statements are constructed using estimates and assumptions*—GAAP allows for considerable room for legitimate reporting differences between firms in similar industries.

2. *Management prepares its own financial statements*—Firms report the results of their activities, and there are considerable opportunities for manipulation. Even if the statements are audited (as all publicly held firms must be), the auditor provides only an opinion as to whether the statements were prepared in accordance with GAAP; the auditor does not prepare them.
3. *Alternative reporting options within GAAP provide the basis for reporting differences*—Within generally accepted accounting principles (GAAP), there are allowances for different methods of reporting a given event or transaction that lead to very different results. For example, inventories can legitimately be accounted for using a last in–first out (LIFO) or a first in–first out (FIFO) method, and fixed assets can be depreciated using one of several methods.
4. *Nonfinancial information contained in the firm's financial statements can be critical*—Firms regularly report information in footnotes and explanatory notes that can be very important when trying to understand and forecast a firm's future performance. For example, assets that are leased with operating leases do not appear in the firm's balance sheet, and future lease payments are not included as liabilities. However, operating lease payments, which can in fact be a significant liability, are reported in the footnotes to the firm's balance sheet.

Performing a Financial Statement Analysis

For business valuation purposes, financial statement analysis provides a tool for learning about the current state of a firm's operations, which can help us avoid mistakes when we are forecasting the firm's future cash flows. Some examples of the types of forecast errors that we want to avoid are discussed in Figure 4.

The set of problems described in Figure 4 arises either because of bad luck or ineffective management of the firm's resources. A careful analysis of a firm's financial statements can also help uncover evidence of earnings manipulation by the firm's management. Earnings are sometimes manipulated by managers to paint a rosier picture of firm performance. In some instances, the manipulation is simply designed to smooth out reported earnings, while in others it is designed to fraudulently disguise the firm's true performance. A very suspicious example of the former is the 100 straight quarters of rising earnings per share at General Electric (GE) under the leadership of CEO Jack Welch. An example of the latter is the case of Worldcom, which increased earnings by capitalizing costs that should have been expensed in the current period.

Step 2: Prepare Pro Forma Financial Statements for the Planning Period

Our earlier analysis showed how the firm's historical financial statements can be recast to reveal the firm's historical free cash flows. Similarly, we can calculate a firm's forecasted free cash flows from pro forma income statements and pro forma balance sheets.

The Forecast Horizon

The value of a firm reflects cash flows that are expected to be generated over many years in the future. However, because forecasting distant cash flows is very difficult, the typical practice in industry is to break the future into two segments: (1) a planning period of three to five years and (2) everything after the end of the planning period. Analysts prepare very detailed projections of revenues, expenses, and cash flows for the planning period. The values of the post-planning-period cash flows are then lumped together in

Figure 4 Problems that can be Detected by Financial Statement Analysis

Problem	Financial Statement Analysis—Detecting the Problem
<i>Uncollectible accounts receivable.</i> Just because the firm has an accounts receivable balance outstanding does not mean that the accounts can be collected in a reasonable period of time.	Compare the firm's <i>average collection period</i> (ACP) to other peer firms and look for trends over time that might suggest deterioration in collection times. An increasing ACP over time is often a red flag that collection problems are growing. An <i>aging schedule</i> of accounts receivable tells the analyst what proportion of the firm's accounts pay on time or are late and provides a way to identify slow-paying accounts.
<i>Unsalable inventory.</i> The firm may have items listed in its inventories that are either not salable or salable only at reduced prices.	Compare the firm's <i>inventory turnover ratio</i> to other peer firms and to itself over time. If the firm's inventory turnover is lower than that of its peers, this could suggest the presence of obsolete or unsalable inventory items. Also, an improving turnover ratio over time may signal that the firm is discounting its merchandise, so a simultaneous review of profit margins is helpful when looking for this problem.
<i>Overstated residual (book) values of property, plant, and equipment.</i> The firm's stock of fixed assets may be outdated or obsolete.	Assess the firm's efficient utilization of fixed assets using the <i>fixed asset turnover ratio</i> . This would entail making peer firm comparisons as well as time series comparisons for the firm itself. Deterioration in this ratio over time and when compared to peer firms suggests either obsolete equipment or inefficient utilization of existing equipment. If a problem is indicated, then a more thorough investigation including technical advice would be called for before proceeding.
<i>Excessive probability of financial distress.</i> The firm's use of debt financing may be such that the probability of default in the event of an economic downturn is excessive in the eyes of the individual or the firm doing the valuation.	Debt utilization can and should be assessed in terms of the level of debt used in the firm's capital structure (i.e., the ratio of debt to assets and debt to equity) as well as the firm's ability to service debt (i.e., using the times interest earned or fixed charge coverage ratio). These ratios should be compared with peer firms and also over time for the same firm so as to detect any trends. In addition, the debt footnote that accompanies the firm's annual 10K filing with the SEC provides information concerning the rate of interest and maturities of the firm's debt as well as any off-balance-sheet financing it may have outstanding. ³

something called the terminal value, which hinges on making an assumption about the rate of growth in cash flows after the end of the planning period. We will illustrate the valuation of post-planning-period cash flows in a terminal value later.

³ For example, the debt footnote to Starbucks Corporation's (SBUX) 2008 10K revealed that the firm's long-term debt was comprised of \$550 million of 6.25% Senior Notes issued in 2007 that are due in August 2017, with interest paid on February 15 and August 15.

Developing a Revenue Forecast

The key to preparing an unbiased estimate of a firm's future cash flows is an unbiased and informed forecast of the firm's future revenues. Revenue forecasts are typically prepared in one of two ways: top down or bottom up. The top-down approach begins with macro- or market-level data and then focuses on market share to get firm revenue estimates. The bottom-up method begins with the firm's own estimates by product line and customers to formulate an estimate of future revenues. Both approaches have their merits and using both approaches in concert provides a useful way to cross-check the reasonableness of the revenue forecast. Regardless of which method is used, this is a difficult problem that is fraught with error. Consequently, the tools of risk analysis are frequently applied to the firm free cash flow estimates, just as they were applied to project cash flows to evaluate the effect of uncertainty on our estimate of firm value.

For our purposes, we will assume that the revenue forecast for Better Buys Inc. is made using a constant annual growth rate. For the planning period, the predicted rate of growth in revenues is 6%. During the post-planning period, we will assume that the firm's cash flows do not grow but remain equal to the firm free cash flow predicted for the end of the planning period (year 2015).

Forecast Methods Used to Construct Pro Forma Statements

In general, the estimates of the individual entries in the firm's pro forma income statements, balance sheets, and cash flow statements depend upon the relationship between each entry and firm revenues. It is common practice to use the common size statements (like the ones found in Tables 1 and 2) to construct the pro forma financial statements. However, a blind use of the constant percent of sales (common size income statement) and percent of total assets (common size balance sheet) will fail to capture any economies of scale effects, because the percentages are constant for all sales levels. Furthermore, if the firm being valued is an acquisition target, then the acquiring firm will want to impose its own assumptions about how the firm will be operated in performing the valuation analysis. For example, if the acquirer feels that the inventory policies of the acquired firm are lax and result in larger-than-necessary investments in inventories, then future cash flow forecasts will assume lower inventory levels.

Pro Forma Income Statements

Table 5 contains pro forma income statements for Better Buys Inc. spanning the five-year period 2011–2015. In our forecast of Better Buys Inc.'s pro forma statements, we make use of four types of forecast methods:

1. *Constant rate of growth*—Sales are assumed to grow at a constant rate of 6% per year over the planning period.
2. *Percent of sales*—Cost of goods sold and variable operating expenses are estimated to be 64.8% and 16.1% of sales, respectively. These percentages are based upon an average of similar ratios calculated over the period 2008–2010 found in Table 1.
3. *Percent of assets, investments, or liabilities for the prior year*—Depreciation expense is 10% of net property, plant, and equipment balance at the close of the previous year; nonoperating income is 10% of the firm's investments at the close of the prior year; and interest expense is 10% of the sum of the firm's short- and long-term debt for the prior year.
4. *Constant*—The fixed cash operating expenses are assumed to remain unchanged over the forecast period. Historically, the firm's fixed cash operating expenses have totaled \$2 million, and we forecast a continuance of this amount over the next five years.

Table 5 Better Buys Inc. Pro Forma Income Statements for 2011–2015

	Forecast Method	Assumption	2011	2012	2013	2014	2015
Revenues	Constant growth rate	6.0%	\$ 39,452,405	\$ 41,819,549	\$ 44,328,722	\$ 46,988,446	\$ 49,807,752
Cost of goods sold	% of sales	64.8%	(25,578,309)	(27,113,008)	(28,739,788)	(30,464,176)	(32,292,026)
Gross profit	Calculated		\$ 13,874,096	\$ 14,706,542	\$ 15,588,934	\$ 16,524,270	\$ 17,515,726
Operating expenses							
Fixed cash operating expenses	Constant	2,000,000	(2,000,000)	(2,000,000)	(2,000,000)	(2,000,000)	(2,000,000)
Variable cash operating expenses	% of sales	16.0%	(6,338,686)	(6,719,008)	(7,122,148)	(7,549,477)	(8,002,446)
Depreciation expense	% of fixed assets (t-1)	10.0%	(1,600,428)	(1,696,453)	(1,798,241)	(1,906,135)	(2,020,503)
Net operating income	Calculated		\$ 3,934,982	\$ 4,291,081	\$ 4,668,545	\$ 5,068,658	\$ 5,492,778
Nonoperating income	% of investments (t-1)	10.0%	300,000	300,000	300,000	300,000	300,000
Earnings before interest and taxes	Calculated		\$ 4,234,982	\$ 4,591,081	\$ 4,968,545	\$ 5,368,658	\$ 5,792,778
Interest expense	% of total liabilities (t-1)	7%	(885,724)	(877,441)	(859,652)	(832,617)	(795,433)
Earnings before taxes	Calculated		\$ 3,349,258	\$ 3,713,639	\$ 4,108,893	\$ 4,536,041	\$ 4,997,345
Taxes	Tax rate × EBT	40.0%	(1,339,703)	(1,485,456)	(1,643,557)	(1,814,416)	(1,998,938)
Net income	Calculated		\$ 2,009,555	\$ 2,228,184	\$ 2,465,336	\$ 2,721,625	\$ 2,998,407
Dividends paid	Policy variable—fixed \$ amount	\$ 200,000	(200,000)	(200,000)	(200,000)	(200,000)	(200,000)
Addition to retained earnings	Calculated		\$ 1,809,555	\$ 2,028,184	\$ 2,265,336	\$ 2,521,625	\$ 2,798,407

Legend:

- t-1 refers to the period just ended or the prior period.
- The pro forma income statement is a forecast of the income statement for a future period, made using assumptions and estimates of the determinants of future revenues and expenses.

The remaining entries in the pro forma income statement are calculated in accordance with the definition of each item (e.g., Gross profit = Revenues – Cost of goods sold).

Pro Forma Balance Sheets

Table 6 contains the pro forma balance sheets for 2011–2015 and indicates steady growth in the firm's total assets from \$42,057,881 in 2011 to \$52,309,675 by the end of the final year of the forecast. There are two types of forecast methods used to construct these statements:

1. *Percent of sales*—For example, all of the current asset accounts as well as net property, plant, and equipment are assumed be a percent of firm revenues for the period. The use of the percent of sales method implies a fixed relationship between firm sales and the line item being forecast. The forecast percent of sales used to construct the pro forma balance sheets is obtained by averaging the annual percentages for the most recent four years (2007–2010).⁴
2. *Policy variables*—The firm's investments (nonoperating assets), long-term liabilities, and common stock entries are all estimated based on a policy decision. Specifically, the investment account is held constant, as is the common stock account. The long-term liabilities account is used to balance the pro forma balance sheets and is allowed to float up or down in accordance with the firm's needs for financing during the period.

Forecasts of assets and liabilities in the pro forma balance sheet are often made using financial ratios other than the percent of sales or assets, as we have done here. The primary advantage of using such an approach is the information conveyed by the particular financial ratio used to predict the asset or liability. For example, we can forecast the firm's accounts receivable balance using an estimate of the average collection period:

$$\frac{\text{Average Collection Period (ACP)}}{\text{Period (ACP)}} = \frac{\text{Accounts Receivable}}{\text{Sales} \div 365 \text{ days}}$$

To forecast the accounts receivable balance for a future period for which we have some ideas about the average collection period and have developed a sales estimate, we can simply substitute these estimates into the ACP relationship found above and solve for accounts receivable. However, this is equivalent to using a percent of sales approach, where we solve the above equation for accounts receivable as follows:

$$\text{Accounts Receivable} = \left(\frac{\text{Average Collection Period (ACP)}}{365 \text{ days}} \right) \times \text{Sales} \quad (4)$$

Note that the “percent of sales” being implicitly used to estimate accounts receivable is simply the ratio of the ACP ÷ 365 days!

Our point here is not to say that using one method or the other is preferred, for they are obviously equivalent. The appropriate focus of the analyst should be on coming up with a better estimate of the percent of sales or, equivalently, the ACP.

⁴ You may note that we are forecasting the entries in the balance sheet using percent of revenues rather than the percent of assets found in the common size balance sheet. These two methods are equivalent as long as the relationship between revenues and assets remains constant over the forecast period. For example, if revenues were twice the firm's total assets, then forecasting accounts payable as 5% of revenues would be equivalent to forecasting them as 10% of assets.

Table 6 Better Buys Inc. Pro Forma Balance Sheets for 2011–2015

Balance Sheets	Forecast Method	Assumption	2011	2012	2013	2014	2015
Cash and marketable securities	% of revenues	2.0%	\$ 749,596	\$ 794,571	\$ 842,246	\$ 892,780	\$ 946,347
Accounts receivable	% of revenues	21.4%	8,442,815	8,949,384	9,486,347	10,055,527	10,658,859
Inventories	% of revenues	32.7%	12,900,936	13,674,993	14,495,492	15,365,222	16,287,135
Current assets	Calculated		\$ 22,093,347	\$ 23,418,948	\$ 24,824,084	\$ 26,313,530	\$ 27,892,341
Net property, plant, & equipment	% of revenues	43.0%	16,964,534	17,982,406	19,061,351	20,205,032	21,417,334
Investments	Policy variable	Constant	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000
Total	Calculated		\$ 42,057,881	\$ 44,401,354	\$ 46,885,435	\$ 49,518,561	\$ 52,309,675
Accounts payable	% of revenues	22.8%	\$ 8,995,148	\$ 9,534,857	\$ 10,106,949	\$ 10,713,366	\$ 11,356,168
Short-term debt	% of revenues	20.3%	8,008,838	8,489,369	8,998,731	9,538,654	10,110,974
Current liabilities	Calculated		\$ 17,003,987	\$ 18,024,226	\$ 19,105,679	\$ 20,252,020	\$ 21,467,141
Long-term liabilities	Policy variable	Balancing term	4,526,035	3,791,376	2,895,797	1,824,669	562,276
Total liabilities	Calculated		\$ 21,530,022	\$ 21,815,602	\$ 22,001,476	\$ 22,076,689	\$ 22,029,417
Deferred income taxes	% of tax liability*	1.0%	\$ 397,166	\$ 426,875	\$ 459,746	\$ 496,035	\$ 536,013
Common stock	Policy variable	Constant	\$ 10,000,000	\$ 10,000,000	\$ 10,000,000	\$ 10,000,000	\$ 10,000,000
Retained earnings	Calculated		10,130,693	12,158,877	14,424,213	16,945,838	19,744,245
Total owners' equity	Calculated		\$ 20,130,693	\$ 22,158,877	\$ 24,424,213	\$ 26,945,838	\$ 29,744,245
Total liabilities and owners' equity	Calculated		\$ 42,057,881	\$ 44,401,354	\$ 46,885,435	\$ 49,518,561	\$ 52,309,675

The pro forma balance sheet is an estimate of the balance sheet for a future period based on assumptions and estimates of the determinants of the firm's balance sheet.

*Deferred taxes (2011) = Deferred taxes (2010) + .01 × Tax liability (2011) and so forth. Note that because deferred taxes increase by 1% of the firm's tax liability for the year, the effective tax rate is 99% of the stated tax rate, or $.99 \times 40\% = 39.6\%$ as long as this relationship holds.

When the Percent of Sales Method Should Not Be Used

The percent of sales method of financial forecasting is not always best, and the analyst should be aware of its limitations. Let's now look at where the percent of sales method does NOT work. We can do this by considering the equation for a percent of sales forecast such as the one presented in Equation 4 above. Looking at Table 4, we see that the Net property, plant, & equipment (PPE) balance for Better Buys Inc. is forecast to equal 43% of the firm's predicted revenues.⁵ This suggests that if Better Buys Inc. expects \$10 million in revenues, the firm would need \$4.3 million in PPE, and if the firm were to expect \$20 million in revenues, it would need twice this amount of PPE, and so forth. However, in reality, Better Buys probably would not acquire PPE at a constant percent of revenues from year to year but would, in a growing environment, overbuild PPE in anticipation of its future needs. In other words, PPE expenditures tend to be lumpy, so assuming a constant percent of revenues for this asset account is not appropriate.

Another problem with the percent of sales method of forecasting arises when there are economies of scale for a particular asset. For example, Better Buys Inc. may need \$10 million in inventories to support a \$30 million level of sales, but this inventory level (given the firm's ability to replenish inventories as needed) may support up to \$50 million in sales. If this is the case, then the percent of sales made up of inventories would be expected to decline as firm sales rose from \$30 to \$50 million.

The key point here is that the percent of sales method of financial forecasting is appropriate *only* if the asset or liability being forecast does indeed vary as a constant percent of revenues and the firm plans to make adjustments to the account balance on a regular basis in response to changing sales.

Step 3: Convert Pro Forma Financial Statements to Cash Flow Forecasts

Table 7 contains the pro forma cash flow statements for Better Buys spanning the period 2011–2015. The growing amount of cash flow from operations over the period allows the firm to retire more long-term debt (the policy variable chosen to balance the firm's finances). However, the free cash flow forecast that we need to value Better Buys is calculated in Table 8. The firm free cash flow (from operations) increases from \$669,762 in 2011 to \$1,187,334 in 2015, while the after-tax nonoperating income remains constant at \$180,000.

Step 4: Estimate the Terminal Value of Firm Free Cash Flows

We estimate the value of the firm's cash flows for all years after the end of the planning period in a lump sum commonly referred to as a terminal value. Alternative approaches can be taken to estimate terminal value. For now, it is important to

⁵ Do not be confused by the fixed \$3 million in investments. These investments are unrelated to the firm's operating assets and do not include capital expenditures, which vary as a fixed percent of sales.

understand that the terminal value represents the present value of all the firm's free cash flows that are expected to be received beginning with one period after the end of the planning period and extending into the indefinite future.

Table 7 Better Buys Inc. Pro Forma Statement of Cash Flow for 2011–2015

	2011	2012	2013	2014	2015
Cash Flows from Operations					
Net income	\$ 2,009,555	\$ 2,228,184	\$ 2,465,336	\$ 2,721,625	\$ 2,998,407
Depreciation	1,600,428	1,696,453	1,798,241	1,906,135	2,020,503
	\$ 3,609,982	\$ 3,924,637	\$ 4,263,577	\$ 4,627,760	\$ 5,018,910
(Increase) Decrease in accounts receivable	(329,018)	(506,569)	(536,963)	(569,181)	(603,332)
(Increase) Decrease in inventory	(618,584)	(774,056)	(820,500)	(869,730)	(921,913)
	\$ (947,602)	\$ (1,280,625)	\$ (1,357,463)	\$ (1,438,910)	\$ (1,525,245)
Increase (Decrease) in accounts payable	\$ 195,048	\$ 539,709	\$ 572,091	\$ 606,417	\$ 642,802
Increase (Decrease) in deferred income taxes	26,794	29,709	32,871	36,288	39,979
<i>Cash provided by operations</i>	\$ 2,884,223	\$ 3,213,430	\$ 3,511,077	\$ 3,831,555	\$ 4,176,446
Cash Flows from Investing Activities					
(Increase) Decrease in net fixed assets	\$ (2,560,684)	\$ (2,714,325)	\$ (2,877,185)	\$ (3,049,816)	\$ (3,232,805)
<i>Cash flows from investing activities</i>	\$ (2,560,684)	\$ (2,714,325)	\$ (2,877,185)	\$ (3,049,816)	\$ (3,232,805)
Cash Flows from Financing Activities					
Increase (Decrease) in short-term debt	\$ (179,397)	\$ 480,530	\$ 509,362	\$ 539,924	\$ 572,319
Increase (Decrease) in long-term liabilities	61,069	(734,659)	(895,579)	(1,071,128)	(1,262,393)
Increase (Decrease) in common stock	—	—	—	—	—
Dividends paid	(200,000)	(200,000)	(200,000)	(200,000)	(200,000)
<i>Cash flows from financing activities</i>	\$ (318,327)	\$ (454,129)	\$ (586,217)	\$ (731,204)	\$ (890,074)
Change in Cash	\$ 5,211	\$ 44,976	\$ 47,674	\$ 50,535	\$ 53,567
The pro forma statement of cash flow is an estimate of the sources and uses of cash for the firm for a future period based on assumptions and estimates of the determinants of those cash flows. The statement can be estimated using information contained in the current period balance sheet, the pro forma balance sheet for the period, and the pro forma income statement for the period.					

Table 8 Better Buys Inc. Free Cash Flow Forecast for 2011–2015

Panel a. Firm Cash Flow (Firm FCF) Calculation	2011	2012	2013	2014	2015
Net operating income	\$ 3,934,982	\$ 4,291,081	\$ 4,668,545	\$ 5,068,658	\$ 5,492,778
Less: Taxes on operating income	(1,573,993)	(1,716,432)	(1,867,418)	(2,027,463)	(2,197,111)
Net operating profit after taxes (NOPAT)	\$ 2,360,989	\$ 2,574,648	\$ 2,801,127	\$ 3,041,195	\$ 3,295,667
Increase (Decrease) deferred taxes	\$ 26,794	\$ 29,709	\$ 32,871	\$ 36,288	\$ 39,979
Plus: Depreciation	1,600,428	1,696,453	1,798,241	1,906,135	2,020,503
Less: CAPEX	(2,560,684)	(2,714,325)	(2,877,185)	(3,049,816)	(3,232,805)
Less: Change in operating net working capital	(757,764)	(785,892)	(833,045)	(883,028)	(936,010)
Firm free cash flow (from operations)	\$ 669,762	\$ 800,593	\$ 922,009	\$ 1,050,774	\$ 1,187,334
Nonoperating income after taxes	180,000	180,000	180,000	180,000	180,000
Free cash flow available for investors	\$ 849,762	\$ 980,593	\$ 1,102,009	\$ 1,230,774	\$ 1,367,334
Panel b. Equity Free Cash Flow (Equity FCF) Calculation	2011	2012	2013	2014	2015
Net income	\$ 2,009,555	\$ 2,228,184	\$ 2,465,336	\$ 2,721,625	\$ 2,998,407
Plus: Depreciation	1,600,428	1,696,453	1,798,241	1,906,135	2,020,503
Increase (Decrease) in long-term liabilities	61,069	(734,659)	(895,579)	(1,071,128)	(1,262,393)
Increase (Decrease) in short-term debt	(179,397)	480,530	509,362	539,924	572,319
Increase (Decrease) in deferred income taxes	26,794	29,709	32,871	36,288	39,979
Less: Change in operating NWC	(757,764)	(785,892)	(833,045)	(883,028)	(936,010)
Less: Capital expenditures (CAPEX)	(2,560,684)	(2,714,325)	(2,877,185)	(3,049,816)	(3,232,805)
Equity free cash flow (Equity FCF)	\$ 200,000	\$ 200,000	\$ 200,000	\$ 200,000	\$ 200,000

4 LOOKING AHEAD—THE MECHANICS OF CALCULATING ENTERPRISE VALUE

Now that we have estimates of the firm's cash flows (both operating and nonoperating), we estimate the enterprise value of the firm as the sum of the present values of its operating cash flows (i.e., the present value of firm free cash flow from operations) plus the present value of nonoperating cash flows, i.e.:

$$\text{Enterprise Value} = \text{Present Value of Firm FCF from Operations} + \text{Value of Nonoperating Cash Flows} \quad (5)$$

Planning Period and Terminal Value

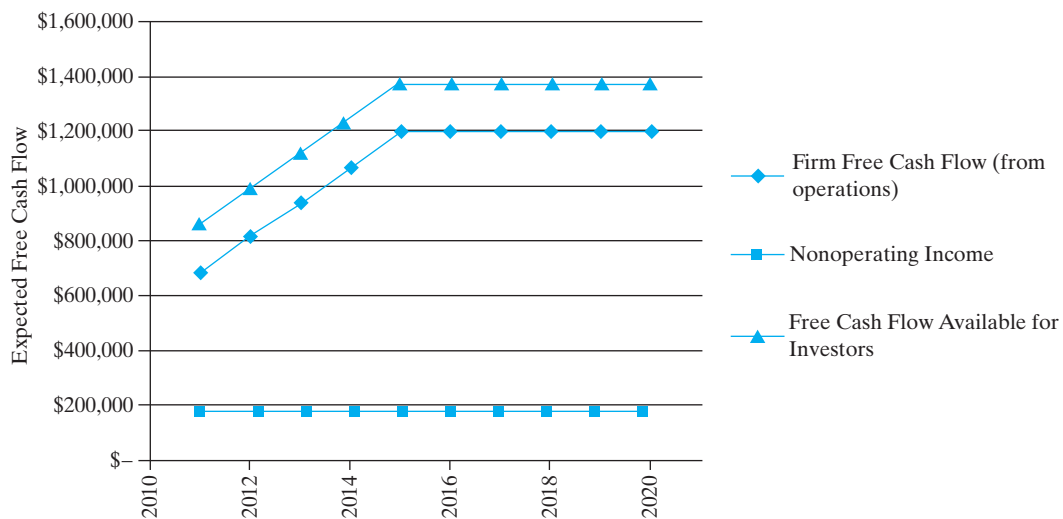
Note that in our previous discussion of forecasting, we forecast only the next five years of cash flows for Better Buys Inc. even though the firm would typically be expected to survive for many years. In theory, because a business has an indefinite and unpredictable life,

the analyst would want to forecast future cash flows over a very long period of time. In practice, however, analysts typically prepare detailed financial forecasts (such as those underlying the preparation of Table 8) for a **planning period** of three to five years. The value of the remaining cash flows for all years following the end of the planning period is then estimated as a **terminal value**. The terminal value can be estimated in a number of ways that we discuss in more detail in section entitled “Enterprise Valuation”; however, for now we will estimate the present value of the post-planning-period cash flows for Better Buys Inc. by assuming that these cash flows remain constant after the end of the planning period such that they are a level perpetuity equal to the firm’s free cash flow for 2015.

Figure 5 describes the time series of cash flows we have estimated for Better Buys spanning the period from 2011 through 2020. Of course, the firm continues to produce cash flows into the indefinite future, which are assumed to be equal to the cash flow realized in 2015. Standard valuation practice requires that we estimate the value of the operating and nonoperating cash flows independently and sum them to estimate the firm’s enterprise value as we described in Equation 5 above.

One final note about nonoperating cash flows is in order. In our Better Buys Inc. example, the nonoperating cash flows are a level perpetuity of \$300,000 per year (\$180,000 after taxes). In practice, nonoperating income often comes from nonrecurring sources such as the sale of assets arising out of restructuring and other extraordinary sources. These cash flows could be much more difficult to predict and take the form of a lump-sum cash flow. Nonetheless, they can potentially be very important sources of value to the firm and must be forecast.

Figure 5 Better Buys Inc. Free Cash Flow Forecast for 2011–2020



Better Buys Inc. is expected to receive a growing stream of free cash flow from its operations, plus a constant after-tax cash flow of \$180,000 from its nonoperating income over the period. Although only 10 years of cash flow are included in the figure, the cash flows are assumed to remain constant into the indefinite future.

However, because we segment the future into a planning period followed by all subsequent cash flows, we modify Equation 5 as follows:

$$\text{Enterprise Value} = \left(\begin{array}{l} \left[\begin{array}{c} \text{Present Value of} \\ \text{Firm FCF from Operations} \\ \text{for 2011–2015} \end{array} \right] + \left[\begin{array}{c} \text{Present Value of} \\ \text{Nonoperating Income} \\ \text{for 2011–2015} \end{array} \right] \\ + \\ \left[\begin{array}{c} \text{Present Value of} \\ \text{Terminal Value for} \\ \text{Firm FCF from Operations} \end{array} \right] + \left[\begin{array}{c} \text{Present Value of} \\ \text{Terminal Value for} \\ \text{Nonoperating Income} \end{array} \right] \end{array} \right) \quad (5a)$$

The estimation of enterprise value using Equation 5a involves the calculation of the value of the planning period cash flows (operating and nonoperating) and adding this to the value of the terminal values (operating and nonoperating).

If the appropriate discount rate for valuing the operating cash flows is 15% and the corresponding discount rate for nonoperating cash flows is 10%, then we can estimate the enterprise value of Better Buys Inc. using Equation 5 as follows:

	Value of Cash Flows		
	Planning Period (2011–2015)	Terminal Value	Total
Firm free cash flow (from operations)	\$ 2,985,098.98	\$ 3,935,430.78	\$ 6,920,529.75
Nonoperating income	682,341.62	1,117,658.38	\$ 1,800,000.00
Total	\$ 3,667,440.59	\$ 5,053,089.16	\$ 8,720,529.75

We estimate the value of Better Buys Inc. at the end of 2010 to be \$8,720,529.75, which represents the value of a stream of free cash flows from operations equal to \$6,920,529.75 and a stream of nonoperating income worth \$1,800,000.00.

There is another approach to the estimation of the value of a business based on market multiples. This methodology is widely used to estimate both enterprise value and project value. Moreover, it is widely used as the basis for estimating terminal values even where it is not used to value planning period cash flows.

5 SUMMARY

Firms publish financial statements that document their earnings and financing histories. Because a firm's ability to generate cash flows in the past provides important insights about its ability to generate future cash flows, valuation experts will benefit from the skills required to unlock the information in these financial statements.

In this chapter, we have reviewed the format of the firm's financial statements and linked them to the firm's free cash flow. This analysis revealed that although financial statements are prepared in accordance with the rules of accrual accounting, the resulting statements can be readily translated back into cash flows that form the basis for valuing the firm.

PROBLEMS

- 1 **CASH FROM OPERATIONS AND FREE CASH FLOW** The item called “Cash provided by operations” in the cash flow statement sounds a lot like firm free cash flow. Although these two quantities are related, they are not the same. Discuss the three sources of difference between them.
- 2 **DEALING WITH NONOPERATING INCOME** What is “other” or “nonoperating income,” and how should it be incorporated into the valuation of a firm?
- 3 **FINANCIAL STATEMENT ANALYSIS** Financial statement analysis can be used to identify weaknesses in a firm’s operations. Give some examples of the types of problems that can be identified and how the financial analyst might uncover them.
- 4 **INCOME VS. CASH FLOW** The guiding principles underlying the accrual basis of accounting drive a wedge between a firm’s reported income and its cash flow. What are some of the reasons for the differences in net income and cash flow?
- 5 **FINANCIAL FORECASTING PROCESS** Describe the four-step financial forecasting process as it is used in the estimation of the value of a firm. How does the percent of sales forecasting method fit into this process?
- 6 **ACCRUAL VS. CASH ACCOUNTING** A firm’s net income is calculated as the difference in the revenues it recognizes for the period less the expenses it incurred in the process of generating those revenues. The method used to determine what revenues are earned and what expenses have been incurred is determined in accord with the basic rules of accrual, not cash, accounting. Consider the following situation:
 - On March 1, 2010, the Sage Video Store in Tacoma, WA, placed an order for 10 52-inch Sony LCD television sets. The vendor they ordered from gave the firm 60 days to pay the \$1,200 cost per set.
 - On April 12, 2010, Mary Griggs came into the Sage Video Store and purchased one of the Sony television sets for \$1,950. Part of Mary’s motivation for making the purchase was that she did not have to make a payment for the purchase for six months and did not have to make a down payment either.
 - a. If Sage ends its first quarter on May 1, how does the sale of the television to Mary impact the firm’s gross profits for the quarter? What is the impact of the sale on the firm’s cash flow?
 - b. If Mary were to make a 10% cash down payment at the time of the purchase and then not begin making payments for six months, what is the impact of the sale on the firm’s gross profit and cash flow?
- 7 **COMMON SIZE FINANCIAL STATEMENTS** The balance sheet and income statement for Webb Enterprises Inc. are found below:

Forecasting Financial Performance

Balance Sheet	2010
Cash and marketable securities	\$ 500
Accounts receivable	6,000
Inventories	9,500
Current assets	<u>\$ 16,000</u>
Net property plant & equipment	17,000
Total	<u><u>\$ 33,000</u></u>
Accounts payable	\$ 7,200
Short-term debt	6,800
Current liabilities	<u>\$ 14,000</u>
Long-term debt	7,000
Total liabilities	<u>\$ 21,000</u>
Total owners' equity	12,000
Total liabilities and owners' equity	<u><u>\$ 33,000</u></u>

Income Statement	2010
Revenues	\$ 30,000
Cost of goods sold	(20,000)
Gross profit	<u>\$ 10,000</u>
Operating expenses	(8,000)
Net operating income	<u>\$ 2,000</u>
Interest expense	(900)
Earnings before taxes	\$ 1,100
Taxes	(400)
Net income	<u><u>\$ 700</u></u>

- a. Prepare a common size balance sheet for Webb Enterprises.
- b. Prepare a common size income statement for Webb Enterprises.
- c. Use your common size financial statements to respond to your boss' request that you write up your assessment of the firm's financial condition. Specifically, write up a brief narrative that responds to the following questions:
 - i. How much cash does Webb have on hand relative to its total assets?
 - ii. What proportion of Webb's assets has the firm financed using short-term debt? Long-term debt?
 - iii. What percent of Webb's revenues does the firm have left over after paying all of its expenses (including taxes)?
 - iv. Describe the relative importance of Webb's major expense categories, including cost of goods sold, operating expenses, and interest expenses.

8 USING COMMON-SIZE FINANCIAL STATEMENTS The B&G Construction Company expects to have total sales next year totaling \$15,000,000. In addition, the firm pays taxes at 35% and will owe \$300,000 in interest expense. Based on last year's operations, the firm's management predicts that its cost of goods sold will be 60% of sales and operating

expenses will total 30% of sales. What is your estimate of the firm's net income (after taxes) for the coming year? What else would you need to know in order to be able to calculate the firm's free cash flow for next year?

9 STATEMENT OF CASH FLOWS The statements of cash flow for Wal-Mart (WMT) are found below for the years 2005 through 2008:

In Millions of USD (except for per share items)	12 Months ended			
	1/31/2008	1/31/2007	1/31/2006	1/31/2005
Net income	12,731.00	11,284.00	11,231.00	10,267.00
Depreciation and non-cash items	6,470.00	6,308.00	4,720.00	4,265.00
Deferred taxes	(8.00)	89.00	(129.00)	263.00
Changes in working capital	1,161.00	2,271.00	2,419.00	249.00
Cash from operating activities	20,354.00	19,952.00	18,241.00	15,044.00
Capital investment expenditures	(15,670.00)	(14,463.00)	(14,186.00)	(12,351.00)
Cash from investing activities	(15,670.00)	(14,463.00)	(14,186.00)	(12,351.00)
Interest and financing cash flows	(334.00)	(227.00)	(349.00)	113.00
Total cash dividends paid	(3,586.00)	(2,802.00)	(2,511.00)	(2,214.00)
Issuance (retirement) of stock, net	(7,691.00)	(1,718.00)	(3,580.00)	(4,549.00)
Issuance (retirement) of debt, net	4,477.00	(92.00)	4,018.00	4,041.00
Cash from financing activities	(7,134.00)	(4,839.00)	(2,422.00)	(2,609.00)
Foreign exchange effects	252.00	97.00	(101.00)	205.00
Net change in cash	(2,198.00)	747.00	1,532.00	289.00

- Describe Wal-Mart's cash flows from operations for the period. Is the firm generating positive cash flow? How has this cash flow changed over time?
- Is Wal-Mart investing cash in its business? How has Wal-Mart been financing its operations?
- How would you describe Wal-Mart's pattern of net cash flows for the period?

10 ESTIMATING CAPITAL EXPENDITURES At the end of fiscal year 2009, Emerson Electric Co. (EMR) had net property, plant, and equipment equal to \$3.5 billion. The ending balance for 2008 was \$3.507 billion, and the firm had depreciation expense during 2009 equal to \$0.727 billion. How much did the company spend on new property, plant, and equipment during 2009?

11 CALCULATING OPERATING NET WORKING CAPITAL Hildebrand Corp. had the following current asset and current liability account balances for 2007 through 2010:

In Millions of \$	2010	2009	2008	2007
Cash & equivalents	\$ 807,926	\$ 560,960	\$ 617,866	\$ 658,255
Short-term investments	178,994	197,408	182,442	614,513
Cash and short-term investments	986,920	758,368	800,308	1,272,768
Accounts receivable	2,028,060	2,365,823	2,341,609	2,302,447
Inventories	1,456,271	1,441,024	1,450,258	1,262,308
Total current assets	4,471,251	4,565,215	4,592,175	4,837,523
Accounts payable	1,138,770	1,601,413	1,584,959	1,416,367
Accrued expenses	878,454	901,546	902,164	863,683
Notes payable/short-term debt	1,038,633	789,285	979,675	821,126
Current port of LT debt	531,635	386,879	303,214	248,028
Total current liabilities	\$ 3,587,492	\$ 3,679,123	\$ 3,770,012	\$ 3,349,204

- Compute Hildebrand's operating net working capital for 2007–2010.
- What are the annual changes in operating net working capital for the sample period?
- Analyze the firm's needs for additional capital to support its need for operating working capital.

12 FORECASTING PRO FORMA FINANCIAL STATEMENTS Prepare a pro forma income statement and balance sheet for Webb Enterprises, found in Problem 7, where revenues are expected to grow by 20% in 2011. Make the following assumptions in making your forecast of the firm's balance sheet for 2011:

- The income statement expenses are a constant percent of revenues except for interest, which remains equal in dollar amount to the 2010 level, and taxes, which equal 40% of earnings before taxes.
- The cash and marketable securities balance remains equal to \$500, and the remaining current asset accounts and fixed assets increase in proportion to revenues for 2010.
- Net property, plant, and equipment increase in proportion to the increase in revenues.
- Accounts payable increase in proportion to firm revenues.
- Owners' equity increases by the amount of firm net income for 2011 (no cash dividends are paid).
- Long-term debt remains unchanged, and short-term debt changes in an amount that balances the balance sheet.

13 FORECASTING FIRM FCF Using your pro forma financial statements from Problem 12, estimate the firm's FCF for 2011.

14 ENTERPRISE VALUE An analyst at the Starr Corp. has estimated that the firm's future cash flows for the next five years will equal \$80 million per year. In addition, the terminal value she estimates for the end of Year 5 is \$400 million.

- a. If the cost of capital for Starr is 12%, what is the value of Starr Corp.'s planning period cash flows spanning the next five years?
- b. It is extremely difficult to estimate cash flows in the distant future, so it is standard practice to lump all the firm's cash flows for years beyond a planning period of, say, five years into something called a terminal value. If companies in Starr's industry are currently selling for five times their annual cash flow, what would you estimate the terminal value of Starr to be in Year 5?
- c. Applying the same discount rate to the terminal value as to the planning period cash flows, estimate the enterprise value of Starr today.

15 ACCRUAL VS. CASH ACCOUNTING In 2010, Warner Manufacturing produced and sold 2 million bulldog trailer-hitch assemblies. The units were produced at a cost of \$45 per unit and sold for \$60 each. However, the cash expenses incurred for each unit only amount to \$20 per unit because the firm has fixed operating costs (including depreciation and other overhead costs that total \$50 million per year). The company is considering a change in its inventory management process that would reduce its average inventory balance down to 100,000 units rather than the 300,000 units it now carries. If the firm institutes the change in 2011, how will the firm's earnings be affected, assuming that the firm simply reduces its production next year by the 200,000-unit reduction in ending inventory? How will the firm's cash flow be impacted by the change in inventory policy?

Detecting Financial Fraud

Figure A-1 provides a list of seven financial shenanigans identified by noted forensic accountant Howard Schilit. The details related to the identification of these shenanigans are beyond the scope of this book; however, the analyst must always be aware of the possibility that the numbers being reported by the firm's management have been massaged. So, let the user beware!

Figure A-1 Earnings Manipulation and Fraud—Howard Schilit's Financial Shenanigans

Type of Manipulation	Techniques Used to Achieve
<i>Shenanigan #1:</i> Recording revenue too soon or of questionable quality	Aggressive Accounting Practices 1. Recording revenues even though future services remain to be provided. 2. Recording revenues before goods are shipped or customer has accepted shipment. 3. Recording revenues where the customer is not obligated to pay. Recording Revenues of Questionable Quality 4. Related party transactions. 5. Providing a "bribe" to the customer to encourage the purchase. 6. Grossing up revenue.
<i>Shenanigan #2:</i> Recording bogus revenue	1. Recording sales that lack economic substance. 2. Recording cash received in lending transactions as revenue. 3. Recording investment income as revenue. 4. Recording supplier rebates tied to future required purchases as revenues. 5. Releasing revenue that was improperly held back before a merger.
<i>(continued)</i>	

Figure A-1 *continued*

<i>Shenanigan #3: Boosting income with one-time gains</i>	1. Boosting profits by selling undervalued assets. 2. Including investment income or gains as part of revenue. 3. Reporting investment income or gains as a reduction in operating expenses. 4. Creating income by reclassifying balance sheet accounts.
<i>Shenanigan #4: Shifting current expenses to a later period—increases current-period earnings (reduces future earnings)</i>	1. Capitalizing normal operating costs rather than expensing them. 2. Changing accounting policies and shifting current expenses to an earlier period. 3. Amortizing costs too slowly. 4. Failing to write down or write off assets whose value is impaired. 5. Reducing asset reserve account balances.
<i>Shenanigan #5: Failing to record or improperly reducing liabilities</i>	1. Failing to record expenses and related liabilities when future obligations remain. 2. Reducing liabilities by changing accounting assumptions. 3. Releasing questionable reserves into income. 4. Creating sham rebates. 5. Recording revenue when cash is received, even though future obligations remain.
<i>Shenanigan #6: Shifting current-period revenue to a later period—reduces current-period earnings (increases future earnings)</i>	1. Creating reserve accounts and releasing them into income in a later period. 2. Holding back revenue just before an acquisition closes.
<i>Shenanigan #7: Shifting future expenses to the current-period as a special charge—decreases current-period earnings (increases future-period earnings)</i>	1. Inflating the amount included in a special charge. 2. Writing off in-process R&D costs from an acquisition. 3. Accelerating discretionary expenses into the current period.

Source: Howard Schilit, *Financial Shenanigans: How to Detect Accounting Gimmicks & Fraud in Financial Reports*, 2nd ed. (New York: McGraw Hill, 2002). Table compiled by authors.

